frame proteins (ORF) 1) vary widely between strains 2) have a wide variety of immune modulating functions/functions that enhance replication but can be removed for in vitro studies and often do not attenuate replication in vitro and 3) have strain specific mechanisms for how immune modulation is achieved. CoV recombination events are relatively common and having the accessory open reading frame genes at the end of the genome makes it easier for recombination events to occur but harder to track how these changes contribute to overall virus fitness. To me (I only know one of the authors) this was written to provide reasonable scientific arguments to quiet the voices that were screaming about the pandemic being caused by a lab released virus. I do appreciate this doesn't make the arguments true but they are plausible.

Combination of multiple scenarios to lead to virus: I agree that the scenarios could happen in combination but still it could have been something that nature cooked up and not a lab with an infectious clone. It could be a combination of all three in theory. Passage adaptation in a lab is artificial at best. Adaptation to Vero cells brings in the issues of selecting viruses in the absence of a key part of the innate immune response. It's part of the reason why so many viruses grow in Vero cells. Bat cell lines are hard to come by and if you passage in these you are driving the virus/bottle necking the quasi species to prefer bat and not human receptors. Even primary cell passages can be tricky as each donor will express different levels of receptor and I've found a surprising number of donors whose cells were not permissive for either SARS-CoV , MER-CoV or both.

Letter to Nature: The letter to Nature reference was the first isolation of replication competent SARS-like viruses from bat and the first to show replication in human cells of the WIV 1 isolate. Another reference that should have been included is from the same group (PMC5708621) and identified the 4231 bat SARS like virus and did work putting different SARS like bat spikes into the WIV 1 infectious clone backbone and demonstrating which were replication competent. Hu et. al. discuss how recombination events likely resulted in the accessory open reading frame genes found in 2003 SARS-CoV. As the group from Wuhan has found most of the SARS-CoV like bat CoV strains in guano from the same cave in China recombination events are the most likely way to change larger chunks of the genome.

I am not sure how the pangolin fits into all of this but given the mixing of animal species in the exotic food markets mixing pangolin poo and bat guano and then people lacking a good sense of hygiene could result in the right set of conditions to get us where we are now.

If this is not useful please let me know and I will try again! Amy

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From: "Wunschel, David S" <David.Wunschel@pnnl.gov>
Date: Thursday, May 14, 2020 at 7:05 PM
To: "Sims, Amy C" <amy.sims@pnnl.gov>
Cc: "Omberg, Kristin M" <kristin.omberg@pnnl.gov>, "Wahl, Karen L" <karen.wahl@pnnl.gov>
Subject: Nature Med Letter

HI Amy,

We just want to follow up with you on a conversation we had earlier in regard to the SARS-CoV-2 virus and some of the observations that people have made about how it differs from other SARS-like corona viruses. I am sure you have been flooded with questions and are more familiar with the debate about the origins of the virus than we are, so we were wondering if you had any thoughts about the arguments put forward by the group from Scripps in the letter to the editor in Nature Med. They propose three hypotheses on the proximal origins that make sense, but focus a lot on the S protein characteristics and don't discuss how the changes elsewhere in the genome from the nearest bat isolate (RatG13 ~ 96-97% similar to SARS-CoV2) could